

EXTRACTION OF STRUCTURED DERIVATIVES FROM INTERTWINED TEXTS

Protocol v0.3 · V3

A Comprehensive Framework for Decomposition with Identity-Anchored Derivatives

A method for extracting modular, queryable, and reusable knowledge units from dense, multi-concept prose while preserving internal coherence and enabling reference-conditional synergistic problem-solving — and, from this version onward, minting every derivative as a permanently identified, addressable node of a continuous knowledge graph.

June 11, 2026

uid: 019eb86f-2711-7d2f-a321-7da8e9fc1e1f {uid:e9fc1e1f}
kind: protocol · version: 0.3 · status: provisional · steward: mohsen-dirbaz {uid:aad26a3e}

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PART I

Foundations: Conceptual Framework & Principles

Compiled on June 11, 2026

Document Specifications

Protocol	General Protocol: Extraction of Structured Derivatives from Intertwined Texts
Version	0.3 — designated V3 in the Identity Integration Roadmap (V3 → V4 → V5, §14)
Supersedes	Version 0.2 (execution date 2026-02-01)
Execution Date	2026-06-11
Review Cycle	Quarterly
Identity	uid: 019eb86f-2711-7d2f-a321-7da8e9fc1e1f {uid:e9fc1e1f} · kind: protocol · status: provisional · steward: mohsen-dirbaz {uid:aad26a3e} · minted_by: claud-fable-reviewer {uid:d90bd5f7}. Registry append pending steward validation (§8.14); versioning follows UNID §8 — versions are events on one anchor.
Normative language	The key words MUST, MUST NOT, SHOULD, and MAY are to be interpreted as in RFC 2119.
Changes from 0.2	See §17 (Version History) for the complete change register.

1 General Overview

This protocol provides an integrated framework for extracting valuable and structured derivatives from complex, intertwined prose. It combines principles derived from conceptual decomposition (separating interwoven threads) into a general method applicable to any intertwined text.

Core Principle

Valuable derivatives maintain the internal coherence of concepts while achieving conceptual separation. They are **extracted, not synthesized** — each element in the output must be traceable to the source.

Functionally, the protocol is an **entropy-reduction engine**: it takes high-entropy narrative texture — concepts fused at the sentence level, dependencies implicit, synergies latent — and resolves it into low-entropy, modular, queryable units. Version 0.3 adds two structural commitments to this engine:

1. **Component-resolved reference interaction.** The reference layer ceases to be a monolith. Every component of every reference-layer format is bound to a defined interaction operator, a phase binding, and a dedicated output obligation (§3.5). Nothing in the reference layer is inert.

2. **Identity at birth.** Every derivative is minted a permanent UNID anchor at the moment of identification (§6.2, Step 1.2). The protocol stops producing isolated flat files and starts producing identity-anchored, addressable nodes of a continuous knowledge graph (§14).

Determining Value

Derivatives are valuable when they maximize functional independence (minimize interdependencies) and reveal synergistic problem-solving potentials conditional on a reference layer.

Narrative Layering

Derivatives are positioned in a three-dimensional angular space relative to a reference layer:

- **Vertical axis:** Abstraction gradient (concrete ← abstract)
- **Horizontal axis:** Analogical distance (direct ← analogical)
- **Diagonal axis:** Domain-specificity gradient (universal ← domain-specific)

Synergistic Problem-Solving Potentials

These arise when derivatives in different angular positions are combined to address challenges present in the reference layer.

Negative Space and the Inverse Protocol

Because this protocol is the entropy-reduction direction of a pair — its inverse is the generative protocol that weaves modular units back into intertwined prose — V3 borrows from the generative direction the one thing a pure extractor structurally lacks: a systematic understanding of **negative space** (what the extraction cannot produce) and **connective tissue** (what the extraction deliberately discards). Both become recorded outputs in V3 (§6.4 Step 3.3, §8.12–§8.13) rather than silent losses. Voids are surveyed and converted into generative seeds for the overall project; connective tissue is ledgered so that the entropy exported out of the units remains auditable — and re-weavable by the inverse protocol.

2 Universal Principles

All derivative extraction methods share these foundational principles.

Definition 2.1 (Preserving Internal Connection)

Each conceptual unit includes four bound components:

- **Claims:** What is stated
- **Mechanisms:** How it works
- **Instances:** Concrete examples
- **Boundaries:** Limitations and caveats

These components must remain bound in the output. An instance separated from its principle loses its meaning. A claim without its mechanism lacks foundation.

Definition 2.2 (Separation Based on Concept, Not Type)

Prohibited: Defining derivatives based on rhetorical type. **Required:** Defining derivatives based on conceptual identity.

Type-based separation cuts the connections that create meaning. Separating components based on rhetorical function rather than conceptual identity compromises coherence and queryability.

Definition 2.3 (Uniform Composition Across Derivatives)

All derivatives must have a similar compositional structure: each includes claims + mechanisms + instances + boundaries (where the source provides them). The composition ratio may vary slightly depending on the concept, but no derivative should be “all X”.

Test: If one derivative includes 90% claims and another includes 90% instances → active failure mode.

Definition 2.4 (Traceability to Source)

Every sentence, phrase, or clause in the output must be traceable to a specific location in the source. You are extracting and reorganizing existing content, not writing new summaries about the content.

Test: Can you cite a source location (paragraph, sentence) for each output element? If not → you have drifted towards re-synthesis.

Definition 2.5 (Functional Independence and Conditional Synergy with Respect to Reference)

Functional Independence: Derivatives should minimize external dependencies. A derivative has high functional independence when it can be understood and applied with minimal reference to other derivatives.

Conditional Synergy with Respect to Reference: The derivative set should reveal synergistic problem-solving potentials relative to a reference layer. Synergistic potentials are not fixed — they are defined by the angular positioning of derivatives relative to the reference layer and to each other.

Optimization Objective: Maximize functional independence within derivatives while maximizing reference-conditional synergistic problem-solving potential between derivatives.

Trade-off: Complete independence eliminates synergistic potential (no common interface exists). Complete intertwining eliminates independence. The protocol seeks the Pareto frontier.

Definition 2.6 (Relevance Primacy) — NEW IN V3, elevated from v0.2 Step 4.8

All derivatives must be built with respect to the reference layer. Content irrelevant to the reference layer is disqualified from presence in a derivative. Derivatives must be traceable to the source, **but traceability cannot waive relevance to the reference**.

This requirement was a Phase-4 verification step in v0.2. In V3 it is a standing invariant that binds every phase: admission of conceptual units (Step 1.1, criterion 6), content assignment (Step 2.1), assembly (Step 3.1), and verification (Step 4.8). V3 additionally makes the invariant **component-resolved**: relevance must be stated against named reference-layer components (a registered challenge RC-n, dimension RD-n, or constraint C-n — §3.5), never generically.

Definition 2.7 (Identity at Birth) — NEW IN V3

The moment a conceptual unit passes the structural filters of Step 1.1, the executing agent **MUST** mint a UNID anchor for it (UUIDv7, with the minimal visa of UNID §6.1) and **MUST** assign it kind: concept. Every extracted derivative is born with a permanent, machine-resolvable identity; cross-links and functional dependencies are written in UNID syntax (`{uid:T8}`) from the first draft onward.

Identity is orthogonal to quality (UNID §8): minting asserts existence and provenance, not merit. Anchors are never deleted, reissued, or reused — a derivative later merged or rejected keeps its anchor and its history.

Definition 2.8 (Conditioning, Not Mixing) — NEW IN V3

Extraction is a conditional operation, not a blend. Derivative **content** originates in the source alone (Definition 2.4); derivative **value** originates in the reference layer alone (Definition 2.6). The process superficially resembles a mixing process — and like a mixing process it can be dominated by whichever input is larger — but the protocol's guarantees hold only while the

two channels remain separate: the source supplies no value judgments, the reference supplies no content.

The Dominance Guard (§3.7) measures and bounds the size asymmetry between the two inputs precisely to protect this conditional independence.

Definition 2.9 (Negative Space Is Output) — NEW IN V3

What the extraction cannot produce (voids: demanded by the reference but absent from the source; gestured at by the source but never developed; or implied by structural mismatch between the reference and derivative dependency graphs) and what it deliberately discards (connective tissue: transitions, framing, and rhetorical scaffolding that bound the concepts in the source) **MUST** be recorded as first-class outputs (§8.12–§8.13). Voids are converted into generative seeds — actionable specifications of what would have to be written or acquired to fill them.

3 Specification of the Reference Layer

3.1 What is the Reference Layer?

The reference layer is the only required input for this protocol in addition to the main source text. This layer defines the conditional context for evaluating the synergistic problem-solving potentials of derivatives. It establishes the reference for derivative positioning and enables the assessment of whether derivatives — individually and in combination — address the challenges present in that reference space.

New in V3: the reference layer is no longer treated as an undifferentiated whole. It is decomposed into its components, and each component is bound to a defined interaction operator, phase binding, and output obligation (§3.5). The reference layer also receives its own entanglement map and dependency DAG (§3.6), and its size relative to the source is governed (§3.7).

3.2 Reference Layer Formats

The reference layer must be specified in one of the following formats. The purpose of these formats is to provide conditional context for judging the value, relevance, and synergy of derivatives; not to predetermine which concepts should be extracted.

Mapping 3.1 (Format 1: Problem Statement)

- **Problem Domain:** [Brief description of the domain] — anchor component, §3.3
- **Challenge:** [Specific problem or question to address]
- **Constraints:** [Relevant limitations — see Constraint Synthesis rule, §3.5]
- **Success Criteria:** [Optional: how solutions will be evaluated]

Mapping 3.2 (Format 2: Contextual Frame)

- **Context:** [Description of the application context] — anchor component, §3.3
- **Stakeholder Perspective:** [Who is asking/using]
- **Key Concerns:** [List of concerns or priorities]
- **Decision Space:** [What decisions or actions are enabled]

Mapping 3.3 (Format 3: Conceptual Anchor)

- **Core Concept:** [Core concept that directs the extraction] — anchor component, §3.3
- **Relevant Dimensions:** [Dimensions against which relevance should be evaluated]
- **Interpretive Lens:** [How to read the source material]

Minimum Reference Format: [Reference: One sentence about the context or problem].

Assumption: if no reference layer is provided, use the stated or implicit purpose of the source document as the reference layer.

3.3 The Anchor Component Invariant — NEW IN V3 (imposed)

Definition 3.1 (Anchor Component Invariant)

The **first component** of every reference-layer format — **Problem Domain** (Format 1), **Context** (Format 2), **Core Concept** (Format 3) — **MUST** be a **condensed form of the raw reference content supplied by the user**. Condensation is performed by deletion and compression only; agent paraphrase is limited to grammatical joining. Every phrase of the anchor component must be traceable to the user-supplied reference material — the reference-side mirror of Definition 2.4.

Purpose. The anchor component is thereby immune to extreme transformations: however aggressively the remaining components are shaped by agent judgment, the original content the user intended to serve as reference is guaranteed to be present, verbatim-traceable, at the origin of the angular space.

Division of authority. The first component, we impose: it is the user's raw content, condensed. All **other** components of the reference layer **SHALL** be determined by the executing agent's judgment (subject to the counter-pattern of §3.4 and the operators of §3.5).

Size. The condensed anchor component **SHOULD** be 50–250 words. **Verification:** Phase 4 includes a Condensation Fidelity audit (Step 4.13) sampling anchor-component phrases against the raw reference.

3.4 Characteristics of a Good Reference Layer

A good reference layer is:

- **Stable:** Does not change during extraction.
- **Specific:** Provides clear context for judging relevance.
- **Actionable:** Enables objective evaluation of problem-solving potential.
- **Unbiased:** Does not predetermine the structure of derivatives.

Counter-pattern: Using the reference layer to predetermine which concepts should be extracted. The reference layer defines **why** extraction is valuable, not **what** should be extracted.

3.5 The Component Interaction Model (CIM) — NEW IN V3

In v0.2 the reference layer interacted with the source **in bulk**: a single relevance test, a single angular origin, and — in practice — improvised challenge lists. V3 replaces bulk interaction with a defined, per-component contract. For every component of every format, the CIM specifies: **(i)** an **interaction operator** — how the component acts on the source during extraction; **(ii)** a **phase binding** — in which phases it acts; and **(iii)** an **output obligation** — its dedicated,

named slot in the final analysis. Nothing in the reference layer is inert; a component with no recorded effect is a verification failure (Step 4.13).

3.5.1 Operator Taxonomy

Operator	Action on the source	Characteristic output
ORIGIN	Fixes the angular origin (0,0,0); all divergence is measured from it. Carried by the condensed anchor component (§3.3).	Origin statement + condensation provenance
SCOPE	Admissibility filter on conceptual units: out-of-domain units are inadmissible regardless of intrinsic quality.	Domain-fit line per derivative; domain coverage statement
TARGET	Mints the enumerated challenge register (RC-1...RC-n) that synergies must address.	RC register + challenge coverage matrix
GATE	Derives the Constraint Ledger (C-1...C-m); every derivative is tested against every constraint.	Constraint Compliance Table (§8.9)
ORACLE	Converts success criteria into Phase-4 checks for the derivative set and its synergies.	Success-criteria scorecard
LENS	Governs the reading stance in identification and the dominance heuristics in assignment.	Lens-application notes
BASIS	Registers named relevance dimensions (RD-1...RD-k); every relevance statement must cite at least one.	Dimension register + dimension × derivative coverage matrix
FRAME	Projects each derivative onto stakeholders: who would use it, and how.	Stakeholder projection line per derivative
SPACE	Maps derivatives and synergies to the decisions they enable.	Decision-enablement table

3.5.2 Format 1 — Problem Statement: Component Bindings

Component	Operator(s)	Phase binding	Output obligation (final analysis)
Problem Domain	ORIGIN + SCOPE	Phases 0–1	Origin statement; domain-fit line per derivative; domain coverage statement. Anchor Component Invariant applies.
Challenge	TARGET	Phases 1, 3, 4	RC register (RC-1...RC-n, each with a citable gloss); challenge coverage

Component	Operator(s)	Phase binding	Output obligation (final analysis)
			matrix; every synergy MUST cite the RC-n it addresses.
Constraints	GATE	Phases 2, 3, 4	Constraint Ledger (C-1...C-m); Constraint Compliance Table (§8.9): every derivative × every constraint, status Honors / Tightens / Violates / Silent, with evidence.
Success Criteria	ORACLE	Phase 4	Success-criteria scorecard: each criterion scored against the derivative set and the synergy catalogue.

3.5.3 Format 2 — Contextual Frame: Component Bindings

Component	Operator(s)	Phase binding	Output obligation (final analysis)
Context	ORIGIN + SCOPE	Phases 0–1	Origin statement; domain-fit line per derivative. Anchor Component Invariant applies.
Stakeholder Perspective	FRAME	Phases 1, 3	Stakeholder projection line in every derivative: who uses it, in what role, at what abstraction level.
Key Concerns	TARGET	Phases 1, 3, 4	Concern register (KC-1...KC-n, functioning as the RC register); concern coverage matrix; synergies cite KC-n.
Decision Space	SPACE	Phases 3, 4	Decision-enablement table: each derivative/synergy mapped to the decision(s) it informs; unmapped decisions are candidate voids.

3.5.4 Format 3 — Conceptual Anchor: Component Bindings

Component	Operator(s)	Phase binding	Output obligation (final analysis)
Core Concept	ORIGIN	Phases 0–4	Origin statement + condensation provenance. Anchor Component Invariant applies.
Relevant Dimensions	BASIS + TARGET	Phases 1–4	Dimension register (RD-1...RD-k); dimension × derivative coverage

Component	Operator(s)	Phase binding	Output obligation (final analysis)
			matrix; every “Relevance to Reference Layer” section MUST cite at least one named RD-n; challenges derived from dimensions enter the RC register.
Interpretive Lens	LENS	Phases 1–2	Lens-application notes: where the lens changed an identification or dominance decision, the decision and the lens clause that drove it are recorded.

Definition 3.2 (Constraint Synthesis Rule — mandatory, format-independent)

Formats 2 and 3 carry no explicit Constraints component. Regardless of format, the executing agent MUST derive a **Constraint Ledger** from the reference content: statements of the form “solutions must...” / “solutions must never...” that the reference imposes on anything claiming to serve it. If no constraints are derivable, the ledger records “none derivable” with justification. The GATE operator then applies to the synthesized ledger exactly as it would to an explicit Constraints component.

Motivating defect. The v0.2-era worked run (source: spontaneous-symmetry-breaking.skill; reference: Adaptive Signal Multiplexer / CQR, Conceptual Anchor format) covered the reference's challenges under an improvised RC-1...RC-6 register — but never addressed its constraints. The reference's binding obligations (latency budgets; impulse-magnitude bounds; memory-monotonicity; conservation tolerance) were tested against no derivative, because no component forced the test. V3 closes this gap by construction.

Worked Illustration — Constraint Ledger the SSB × ASM/CQR run would have produced

- **C-1 (Latency structure):** any mechanism mapped onto the reference must respect the three-scale ladder — Gate <1 ms (ns-scale reads), TB ~100 ms, MB ~1 s+; topology computation stays off the hard real-time path.
- **C-2 (Impulse bound):** impulsive reallocation is admissible only while $|\Delta w|$ respects the rate-clamp bound (reference Obligation O-7); an unbounded nudge violates non-expansiveness.
- **C-3 (Memory-monotonicity):** tightening is free; loosening only through the trusted scalar ϕ and signed updates — any derivative advocating symmetric relaxation conflicts.
- **C-4 (Conservation tolerance):** $\sum w = M$ within tolerance, else fail-closed freeze; derivatives proposing redistribution must preserve the invariant.
- **C-5 (Epistemic discipline):** claims carry warrant / defeater / standing; finite-N caveat

— no thermodynamic-limit protection may be assumed.

Under V3, derivative D4 (Critical Point & Engineered Transition) would have been tested against C-3 — its hysteresis claim aligns with, but was never checked against, the tighten-fast / loosen-after-dwell asymmetry; D5 (Selection vs. Commitment) would have been tested against C-2 — its “infinitesimal nudge” maps to $\Delta w \cdot \delta(t-t_k)$ only under the rate-clamp bound; and every derivative would carry a C-1 row, where the v0.2 run was silent.

3.6 Reference-Side Entanglement Map and Dependency DAG — NEW IN V3

v0.2 produced an entanglement map and a dependency DAG **for the source only**. V3 **REQUIRES** the same two artifacts for the reference layer, at component granularity:

- **Nodes:** the reference-layer components and the registered entities they mint (RC-n challenges, C-n constraints, RD-n dimensions).
- **Edges:** functional dependencies among them — “challenge RC-3 presupposes constraint C-2”; “dimension RD-1 grounds challenges RC-1 and RC-4”.

The construction is cost-effective — after condensation the operative reference is small — and it raises the analytic yield of the whole task in three ways. First, it exposes which challenges presuppose which constraints, so coverage claims can be checked transitively. Second, it enables **homology analysis**: comparing the reference-side DAG with the derivative-side DAG (§8.5) identifies matched edges (structural alignment between what the reference needs and what the derivatives supply) and unmatched reference edges, which are candidate structural voids (§6.4, Step 3.3). Third, it is the graph substrate on which V4's identity propagation operates (§14.2). Both artifacts **MUST** be included in the final analysis (§8.6).

3.7 Mixing Ratio and the Dominance Guard — NEW IN V3

v0.2 counted on the user's discretion to share references of reasonable size with respect to the source. The risk is real: the process has similarities to a mixing process in which either input can dominate. A dominant reference turns extraction into reference-driven cherry-picking — the counter-pattern of §3.4 — while a dominant source starves the relevance gate and produces generic, unconditioned derivatives. V3 makes the balance measured, reported, and governed.

Definition 3.3 (Mixing Ratio and Dominance Bands)

Let $p_{\text{raw}} = (\text{raw reference word count}) / (\text{source word count})$, and $p_{\text{op}} = (\text{operative reference word count}) / (\text{source word count})$, where the **operative reference** is the formatted specification actually conditioning the run (post-condensation), not the raw reference document. Both ratios **MUST** be reported in the Output Header (§8.2).

Band	p_{op}	Disposition
Source-dominant	$p_{\text{op}} < 0.02$	Relevance gate at risk of starvation: enrich the reference specification via the CIM before

Band	ρ_{op}	Disposition
		extraction, or justify proceeding.
Balanced	$0.02 \leq \rho_{op} \leq 0.5$	Proceed.
Reference-dominant	$\rho_{op} > 0.5$	Cherry-picking risk: condensation becomes mandatory normalization. Re-condense; quote sparingly in non-anchor components; link the raw reference by UNID for provenance instead of inlining it.

When ρ_{raw} is large (the raw reference dwarfs the source), the Anchor Component Invariant (§3.3) is the normalization instrument: the condensed anchor caps the active reference mass while the raw document remains addressable — registered (or minted) as an artifact and linked by `<uid>` — for provenance and audit.

Worked Illustration — mixing in the SSB × ASM/CQR run

The raw reference (the ASM/CQR write-up) exceeded the source skill several-fold — a strongly reference-dominant ρ_{raw} . The run survived because the assessment informally condensed the reference into a one-page Conceptual Anchor before extracting: an implicit normalization. V3 makes exactly that move explicit, bounded (50–250 words for the anchor component), measured (ρ_{raw} and ρ_{op} in the header), and auditable (condensation fidelity, Step 4.13).

4 Angular Positioning: Narrative Layering

4.1 Three-Dimensional Divergence Space

Each derivative occupies a position in a three-dimensional space defined by three axes of divergence from the reference layer.

Definition 4.1 (Abstraction Gradient — Vertical). Range: Concrete (0) ← Abstract (1)

- **Concrete (0.0–0.3):** specific instances, tangible objects, named entities, measurable quantities.
- **Intermediate (0.4–0.6):** balances concrete examples and generalizable principles.
- **Abstract (0.7–1.0):** universal principles, formal structures, conceptual frameworks.

Concrete derivatives provide practical contextualization; abstract derivatives provide transferable patterns; combinations enable the embodiment of principles.

Definition 4.2 (Analogical Distance — Horizontal). Range: Direct (0) ← Analogical (1)

- **Direct (0.0–0.3):** explicitly discusses the reference layer's domain, in its terminology.
- **Intermediate (0.4–0.6):** bridges the reference layer and adjacent domains; partial metaphor.
- **Analogical (0.7–1.0):** discusses a different domain; relationship to the reference is through analogy.

Direct derivatives provide immediate applicability; analogical derivatives provide innovative perspectives and constraint relaxation; combinations enable analogical transfer and cross-domain innovation.

Definition 4.3 (Domain-Specificity Gradient — Diagonal). Range: Universal (0) ← Domain-Specific (1)

- **Universal (0–0.3):** applies across all domains; lacks domain-specific vocabulary.
- **Intermediate (0.4–0.6):** applies to a family of related domains.
- **Domain-Specific (0.7–1.0):** deeply embedded in the ontology, methods, and constraints of one domain.

Universal derivatives provide the broadest reusability; domain-specific derivatives provide the highest fidelity to domain constraints; combinations enable contextual adaptation of universal principles.

4.2 Angular Coordinates

Each derivative is assigned a triplet (α, β, γ) : α = abstraction score [0,1]; β = analogical distance score [0,1]; γ = domain-specificity score [0,1].

Positioning procedure: (1) read the derivative's content; (2) score each axis against the measurement criteria; (3) document the score and its rationale; (4) plot in three-dimensional angular space relative to the reference layer.

Reference layer position: conventionally always at the origin (0,0,0) — all divergences are measured relative to it. **V3 clarification:** the origin is carried by the ORIGIN operator — that is, by the condensed anchor component of §3.3. Because the anchor component is verbatim-traceable user content, the origin of the angular space is guaranteed to be the user's intended reference, not an agent reconstruction of it.

4.3 Layering Patterns

Mapping 4.1 (Pattern 1: Vertical Stack — β, γ constant; α variable)

Derivatives share domain and directness but differ in abstraction level. **Synergy:** abstraction laddering — moving from concrete instances to general principles. **Problem-solving:** generalization from specific cases; instantiation of abstract principles.

Mapping 4.2 (Pattern 2: Horizontal Spread — α, γ constant; β variable)

Derivatives share abstraction level and domain-specificity but differ in analogical distance. **Synergy:** analogical reasoning — leveraging adjacent or distant solutions. **Problem-solving:** constraint relaxation and perspective shifts.

Mapping 4.3 (Pattern 3: Diagonal Gradient — α, β constant; γ variable)

Derivatives share abstraction and directness but differ in domain-specificity. **Synergy:** contextual adaptation — applying universal patterns with fidelity to the domain. **Problem-solving:** transfer of methods across domain boundaries with appropriate specialization.

Mapping 4.4 (Pattern 4: Full Spread — α, β, γ variable)

Derivatives occupy diverse positions across all three axes. **Synergy:** multi-faceted problem-solving — combining concrete/abstract, direct/analogical, universal/specific. **Problem-solving:** maximum coverage of problem-solving strategies.

4.4 Synergistic Problem-Solving Potentials

Definition 4.4 (Synergistic Problem-Solving Potential)

A synergistic problem-solving potential exists when two or more derivatives, located at distinct angular coordinates, can be combined to address a challenge in the reference layer that none of the derivatives addresses alone.

Evaluation procedure: (1) identify reference-layer challenges — in V3 these are the **registered** RC-n / KC-n entries minted by the TARGET operator, not ad-hoc readings; (2) assess each derivative's individual contribution; (3) identify angular complementarity; (4) assess the combined contribution; (5) document the synergy with its angular coordinates and mechanism, **citing the RC-n addressed and the participating derivatives by {uid:T8}**.

Angular pattern	Synergy mechanism	Problem-solving capability
α low + α high (Abstraction spread)	Ground abstract principles in concrete instances	Validate generalizability; instantiate framings
β low + β high (Analogical spread)	Transfer solutions across domain boundaries	Apply distant analogies; relax constraints
γ low + γ high (Specificity spread)	Adapt universal patterns to domain constraints	Contextualize general principles; extract domain patterns
Triplet (3+ derivatives, diverse angles)	Multi-perspective triangulation	Robust solutions; multiple validation paths
Adjacent (small Euclidean distance)	Incremental refinement	Detail improvement; nuance differentiation
Opposing (large Euclidean distance)	Maximal contrast	Identify assumptions; explore design-space boundaries

5 When to Use This Protocol

5.1 Diagnostic Questions

Use this protocol when:

- The text addresses several distinct concepts, intertwined at the paragraph or sentence level (not cleanly sectioned).
- You need to query, compare, or reuse concepts independently.
- The text is dense — the information-to-word ratio is high — and narrative flow prioritizes readability over thematic grouping.
- Functional dependencies exist but are not explicit.
- You have a reference layer (problem, context, or goal) against which derivative value should be evaluated.
- Synergistic problem-solving capabilities beyond simple concatenation are needed.
- The derivatives are intended to live on — as addressable nodes available to later extraction passes, recursive reference layers, and chain analysis (V3 identity seeding makes this the default).

Contraindication: this protocol is not needed — simple sectioning is sufficient — if the text is already well-structured with clear sections per concept.

5.2 Value Proposition

This protocol transforms:

- Intertwined prose (readable but unqueryable) → modular, comparable, reusable structured knowledge.
- Linear narrative (one reading path) → multi-dimensional access (query by concept, angular position, and — new in V3 — permanent identity).
- Author's mental model (implicit structure) → explicit structure (formalized relationships in {uid} syntax).
- Hidden dependencies → explicit functional dependencies, on both the source side and — new in V3 — the reference side.
- Latent synergies → reference-conditional synergistic problem-solving potentials with angular positioning.
- Silent losses → recorded negative space: connective-tissue ledger, void map, generative seeds (new in V3).

5.3 Automatic Rejection Criteria

Files are automatically rejected (without gate evaluation) when:

1. **Missing provenance:** the source lacks the identification and provenance components required by the Output Header (§8.2) — including a resolvable or mintable identity for

source and reference. (Editorial repair: v0.2 cited a non-existent §3.5 “evidence package” here; the requirement now points to §8.2 and §8.14.)

2. **Origin falsification:** the source is not identifiable or verifiable.

Content-quality rejection remains orthogonal to identity (UNID §8): a rejected file keeps its anchor — rejection is a verdict about content, not an erasure of identity.

PART II

The Extraction Process

Compiled on June 11, 2026

6 General Process

6.1 Phase 0: Define Reference Layer and Instantiate the CIM

Input: raw reference material supplied by the user, plus a chosen format (§3.2). **Output:** a stable, component-resolved reference-layer specification that conditions all subsequent positioning, relevance, constraint, and synergy evaluations.

Phase 0 Steps

1. **Step 0.1 — Capture and identify the inputs.** Resolve the UNIDs of the source document and the raw reference; if either is unregistered, mint it (kind: artifact) with the minimal visa. Record both in the Output Header (§8.2).
2. **Step 0.2 — Condense the anchor component.** Produce the first component (Problem Domain / Context / Core Concept) by condensation of the raw user-supplied reference per the Anchor Component Invariant (§3.3). Record condensation provenance (which raw passages were kept).
3. **Step 0.3 — Complete the remaining components.** Determine all other components by agent judgment, honoring the counter-pattern of §3.4 (the reference defines why, not what).
4. **Step 0.4 — Instantiate the CIM.** Bind each component to its operators per §3.5; mint the registers: RC register (TARGET), Constraint Ledger (GATE — mandatory in every format per Definition 3.2), dimension register (BASIS), and any FRAME / SPACE / ORACLE structures the format provides.
5. **Step 0.5 — Map the reference side.** Build the reference-side entanglement map and component dependency DAG (§3.6).
6. **Step 0.6 — Apply the Dominance Guard.** Compute p_{raw} and p_{op} , classify the band, and apply the disposition of Definition 3.3 before proceeding.

Phase 0 Example (Format 1: Problem Statement)

Problem Domain (anchor, condensed from user text): control process for multiphase chemical reactors. **Challenge:** reduce oscillatory behavior during startup without sacrificing steady-state yield → RC-1. **Constraints:** real-time loop <100 ms (C-1); limited sensor suite (C-2). **Success Criteria:** startup time <50% of baseline (S-1); steady-state performance ≥95% of baseline (S-2).

CIM instantiation: Problem Domain = ORIGIN + SCOPE; Challenge mints RC-1; Constraints mint C-1, C-2 (GATE); Success Criteria mint S-1, S-2 (ORACLE). Reference-side DAG: RC-1 presupposes C-1 (any anti-oscillation mechanism must fit the loop budget); S-1 depends on RC-1.

6.2 Phase 1: Identification

Mapping 6.1 (Step 1.1: Identify Conceptual Units)

Scan the text and identify distinct conceptual units — ideas, mechanisms, entities, or principles that the source treats as coherent wholes.

Criteria for a valuable conceptual unit:

1. **Dedicated treatment:** not merely a passing reference.
2. **Component completeness:** captures all four component types (claims, mechanisms, instances, boundaries).
3. **Naming clarity:** can be named concisely (2–5 words).
4. **Coherent independence:** could stand alone as a mini-document.
5. **Functional independence:** minimizes dependencies on other units for understanding.
6. **Relevance to reference layer (component-resolved):** addresses at least one **named** register entry — an RC-n / KC-n challenge, RD-n dimension, or C-n constraint. Generic relevance is insufficient (Definition 2.6).

Output: a list of N conceptual units (typically 3–7 for medium-length text). **Naming convention:** use the source's own terminology where possible; otherwise the most prominent noun phrase.

Mapping 6.2 (Step 1.2: Mint at Extraction) — NEW IN V3

The moment a conceptual unit passes the Step 1.1 filters, the executing agent **MUST** mint a UNID anchor for it. Minting is not deferred to assembly or output: the unit is born identified. (UNID §10: admission is cheaper than a TODO.)

1. **Minimal visa generation.** The agent automatically generates the UNID minimal visa alongside the unit's structural components: uid (freshly minted UUIDv7), kind, handles (the concept name, plus one filename-safe ASCII slug), gloss (≤ 140 characters), steward, status: provisional, minted, minted_by, events.
2. **Kind mapping.** The agent **MUST** assign **kind: concept** to every extracted derivative. No other kind is assigned by this protocol in V3; new kinds (synergy, relationship) await the V4 taxonomy amendment (§14.2).
3. **Short anchors.** The tail-8 rule applies (last 8 hex digits, hyphens stripped); head-prefix abbreviation is prohibited. On a tail-8 collision within the report, widen to tail-12 per UNID §4.3.
4. **Accumulation.** Visas accumulate into the Registry Submission Block (§8.14). Direct registry appends are out of scope until V5 (§14.3); the agent **MUST NOT** represent minted-but-unappended records as registry residents.

Worked Illustration — minimal visa minted at extraction (illustrative)

```

uid:    019eb86f-2711-79ad-841b-e79456b33eba # tail-8: 56b33eba
kind:   concept
handles: [the-order-parameter, "D3 — The Order Parameter"]
gloss:   Scalar zero in the symmetric state, growing with commitment;
         measures how decisively symmetry has broken and detects drift back.
steward: <full anchor of mohsen-dirbaz {uid:aad26a3e}>
status:  provisional
minted:  2026-06-11
minted_by: <full anchor of claudie-fable-reviewer {uid:d90bd5f7}>
events:  [{t: 2026-06-11, e: minted, by: <minted_by anchor>}]

```

Illustrative only — drawn from the SSB worked run (derivative D3). Machine surfaces carry full 36-character anchors throughout — including the steward and minted_by fields, whose full anchors are resolved from the registry; tail-8 forms ({uid:56b33eba}) are human-surface only (UNID §5.3).

Mapping 6.3 (Step 1.3: Assess Entanglement Density)

Density	Characteristic	Extraction difficulty
Low	Concepts discussed sequentially; section breaks align with concepts.	Easy — derivatives map to sections.
Medium	Concepts intermingled but separable at the paragraph level.	Medium — requires clause-level assignment.
High	Concepts fused at the sentence/clause level; share instances and mechanisms.	Hard — requires decomposition and cross-referencing.

High-density texts require more careful connection verification, cross-linking, and dependency analysis to maintain functional independence.

Mapping 6.4 (Step 1.4: Map Entanglement Points, Dependencies, and Angular Positions)

Identify where concepts intersect, list their relationships, and perform initial angular positioning. Relationship types and management:

Relationship type	Strategy / independence / synergy notes
Shared instance	Assign to primary concept; note secondary link. Independence: low impact. Synergy: potential through common grounding.
Bridging claim	Include in both derivatives with cross-reference. Independence: medium impact — creates interface. Synergy: high, via explicit connection.
Functional	Document dependency; include minimal prerequisite summary or

Relationship type	Strategy / independence / synergy notes
dependency	cross-reference. Independence: high impact — must be preserved. Synergy: enables compositional reasoning.
Nested mechanism	Note dependency and include. Independence: high impact — asymmetric dependency. Synergy: hierarchical combination opportunity.
Compound sentence	Decompose; assign clauses to respective derivatives. Independence: neutral. Synergy: minimal.
Contextual dependency	Note order constraint. Independence: high impact — requires ordering. Synergy: sequential combination opportunity.
Synergy pair	Document synergy potential in both derivatives. Independence: neutral. Synergy: explicit annotation.

Functional dependency: concept A is functionally dependent on concept B if understanding A requires prior understanding of B. From V3 onward, all dependency and cross-link records use {uid:T8} syntax from the moment units are minted (Step 1.2).

Initial angular positioning: α = (abstract elements)/(total elements); β = 1 – (terminology overlap ratio with the reference layer); γ = (domain-specific elements)/(total elements). Output per unit: initial (α , β , γ) with brief rationale.

Phase outputs: (1) source entanglement map; (2) source dependency DAG; (3) angular positioning diagram; (4) synergy potential matrix — all using {uid} references.

6.3 Phase 2: Assignment

Mapping 6.5 (Step 2.1: Assign Content to Conceptual Units)

For each content element (sentence, clause, phrase), determine which conceptual unit it belongs to.

1. **Primary assignment:** each element belongs to exactly one conceptual unit.
2. **At entanglement points:** assign to the dominant concept; note the secondary link.
3. **Truly general content:** assign to common ground (§7).
4. **Functional prerequisites:** when A depends on B, include a minimal summary of B in A or give an explicit {uid} reference.
5. **Angular positioning evidence:** mark content that determines angular position (abstraction level, analogical markers, domain-specificity indicators).
6. **Constraint evidence (NEW):** flag content that bears on any C-n in the Constraint Ledger — it feeds the GATE test in Phase 3/4.
7. **Blind-spot evidence (NEW):** mark content revealing what the unit does **not** address (explicit boundaries, deferred questions, unengaged register entries) — it grounds the

Blind Spots section in assembly.

Dominance heuristics: Which concept does this element introduce or elaborate? If removed from concept A, would A lose coherence ($\rightarrow$ A dominant)? If the element serves both equally $\rightarrow$ bridging element; include in both with cross-reference. The LENS operator's reading stance (§3.5) governs ambiguous dominance calls; such calls are recorded in the lens-application notes.

Dependency resolution: if A refers to B, include enough of B in A to maintain functional independence, or make the dependency explicit. Prefer explicit dependencies over implicit ones.

Mapping 6.6 (Step 2.2: Verify Component Balance, Dependency Integrity, and Angular Certainty)

For each conceptual unit, verify the assigned content includes: at least one claim; at least one mechanism (or clear cross-reference); at least one instance (if the source provides one); boundaries (if provided); documented functional dependencies (if any); sufficient angular positioning evidence; and **(NEW)** any constraint-relevant and blind-spot evidence flagged in Step 2.1.

If a unit lacks a component type that the source discussed for that concept, content was misassigned — return to Step 2.1. **Dependency integrity:** for each $A \rightarrow B$, either a sufficient summary of B is included in A, or an explicit $\langle \text{uid} \rangle$ cross-reference exists. **Angular certainty:** content provides evidence for all three axes; mark for review where positioning is critical.

6.4 Phase 3: Assembly

Mapping 6.7 (Step 3.1: Structure Each Derivative — V3 template)

Organize each conceptual unit's content into the standard structure. Fields marked • are new or upgraded in V3; the full writing template appears in §8.11.

- **Derivative:** [Concept Name] — $\langle \text{uid:T8} \rangle$ • (full anchor + tail-8 mandatory)
- **Anchor / Kind:** full 36-character uid; kind: concept •
- **Angular Position:** (α , β , γ) with per-axis rationale
- **Claims:** what is stated
- **Mechanism:** how it works
- **Instances:** each tagged “ $\leftarrow$ Illustrates: [claim/mechanism]”
- **Boundaries:** limitations and caveats
- **Functional Dependencies:** $\langle \text{uid:T8} \rangle$ syntax mandatory •
- **Relevance to Reference Layer:** must cite named register entries (RC-n / RD-n / KC-n) •
- **Constraint Compliance:** status against every C-n — Honors / Tightens / Violates /

- Silent, with evidence •
- **Blind Spots:** BS-n entries, grounded per Definition 6.1 •
- **Coverage Means:** per BS-n, the covering instrument per Definition 6.2 •
- **Synergistic Problem-Solving Potentials:** pattern; mechanism; **Addresses:** RC-n; “alone, neither addresses”; partners by {uid:T8}
- **Cross-links:** {uid:T8} only •

Style guide: use the source's language as much as possible; compress only to remove redundancy; never create new content.

Definition 6.1 (Blind Spots — admissible grounds) — NEW IN V3

A blind spot is a systematic limitation of a derivative's field of view: something the derivative cannot see, test, or decide from within itself. Blind spots **MUST** be grounded in one of four admissible sources — invented criticism is a failure mode (§9.19):

1. **Source-stated boundaries:** limitations the source itself declares (the Boundaries component, read as epistemic perimeter).
2. **Unengaged register entries:** reference components the derivative is Silent on — empty cells in its constraint row or coverage columns.
3. **Angular position implications:** high- β derivatives are blind to direct domain mechanics; low- α derivatives to generalization; high- γ derivatives to cross-domain transfer; low- γ to domain fidelity. The position itself predicts the blindness; the content confirms it.
4. **Dependency truncation:** what was cross-referenced away — a derivative that delegates its mechanism to {uid:X} cannot internally validate that mechanism.

Definition 6.2 (Coverage Means) — NEW IN V3

Every blind spot BS-n **MUST** carry a coverage disposition — the means by which it can be covered:

1. **Sibling derivative:** another derivative {uid:T8} in this extraction whose field of view includes the blind spot.
2. **Documented synergy:** a registered combination whose joint field of view covers it (cite the synergy and its RC-n).
3. **Void / generative seed:** no current coverage — the blind spot is promoted to the Void Map and seeds future work (§8.13).
4. **External acquisition:** coverage exists outside this extraction (a named document, dataset, or expertise; by {uid} if registered).
5. **Declared irreducible:** no coverage is possible in principle; the declaration itself is the disposition.

Set-level verification (Step 4.10) assembles the Blind Spot & Coverage Matrix (§8.12) and checks mutual coverage: the derivative set should, where possible, cover its own blind spots — uncovered spots flow to the Void Map by construction.

Worked Illustration — blind spot with coverage (SSB run, derivative D3)

BS-1 (grounds: angular + unengaged C-1): D3 (The Order Parameter {uid:56b33eba}) assumes a scalar readout exists but is silent on **who reads it and at what latency** — untested against C-1 (Gate ns / TB ~100 ms / MB ~1 s+). **Coverage:** sibling D4 (hysteretic transition discipline) plus external acquisition: the reference's latched witness-bit mechanism (S-bit read ~8 ns, computation off the hard real-time path).

Mapping 6.8 (Step 3.2: Add “Articulation Dressing”)

When entanglement forces separation of bound elements, add minimal connecting context:

Situation	Compensation
Instance separated from principle	Bridging phrase: “Illustrates: X”
Claim dependent on mechanism elsewhere	Cross-reference: “Mechanism: see {uid:T8}”
Boundary separated from claim	Minimal claim summary: “Constrains: claim X”
Mechanism refers to another concept's mechanism	Dependency note: “Requires: Y from {uid:T8}”
Functional dependency exists but source concept absent	Minimal prerequisite summary (with assignment)
Angular evidence scattered	Note ambiguity: “α uncertain; abstract markers limited”
Relevance to reference indirect	Make explicit: “Addresses RC-n through Y”

Budget: articulation text $\leq 10\%$ of derivative length. **Anti-pattern:** writing explanatory essays — articulation is brief annotation, not new prose.

Mapping 6.9 (Step 3.3: Negative Space Assembly) — NEW IN V3

After the derivatives and common ground are assembled, account for everything extraction did not capture. This step imports the inverse (generative) protocol's two structural insights.

(a) Connective Tissue Ledger. Narrative material assigned to no derivative and no common ground — transitions, framing moves, rhetorical scaffolding — is not discarded silently. Each load-bearing fragment is ledgered with the relation it encoded: what it joined, what dependency, contrast, or sequence it implied. Rationale: the extractor reduces entropy

inside the units precisely by exporting it into the tissue between them; the ledger keeps the exported entropy auditable, frequently surfaces implicit dependencies that unit extraction loses, and is exactly the material the generative protocol needs to re-weave derivatives into prose.

(b) Void Mapping. Survey the negative space in three classes:

- **Demand voids:** registered reference entries (RC-n, C-n, RD-n, decisions in the SPACE table) that no derivative and no synergy engages.
- **Supply voids:** concepts the source names or gestures at but never develops to Step 1.1 standards (dedicated treatment absent).
- **Structural voids:** mismatches from DAG homology — reference-side edges with no derivative-side counterpart, broken transitive chains, and blind spots whose coverage disposition is “void”.

(c) Generative Seeds. Each void V-n yields a seed specification: void id and class; what would have to be written or acquired to fill it; the register entries it would serve; its candidate angular position; and a candidate handle. Seeds are charters, not anchors: per UNID A5 (no fabricated identity), a seed is NOT minted unless the steward deliberately charters the entity — seeds default to unminted. The seed catalogue is a project-level output: it defines the discovery frontier for subsequent extraction passes and is the designated input of the V4+ Missing Link Predictor (§16.3).

6.5 Phase 4: Verification

Mapping 6.10 (Step 4.1: Check Connection Integrity)

For each derivative: instances are bound (each accompanied by its principle or a cross-reference); claims are grounded (each with its mechanism or cross-reference); boundaries are attached (each constrains a claim present in the same derivative). Any failure → return to Phase 2 and reassign.

Mapping 6.11 (Step 4.2: Check Composition Uniformity)

Verify similar compositional structure across derivatives (claims / mechanisms / instances / boundaries present where the source provides them). **Test:** any derivative >80% one component type → active failure mode.

Mapping 6.12 (Step 4.3: Traceability Audit)

Randomly sample 20% of output elements; verify each is traceable to a specific source location. Failure → re-synthesis has occurred; discard and restart.

Mapping 6.13 (Step 4.4: Check Distinctness)

Content overlap between derivatives <20%; existing overlap occurs at documented entanglement points. Significant overlap → possibly one concept mistakenly split; consider merging (and record the merge in the visas' event history).

Mapping 6.14 (Step 4.5: Assess Functional Independence)

Score range	Interpretation	Action
>80%	High independence — self-sufficient.	No action needed.
60–80%	Medium independence — acceptable with documented dependencies.	Verify dependencies are documented (<code>{uid}</code> syntax).
<60%	Low independence — overly intertwined.	Return to Phase 2; reassign or merge.

Mapping 6.15 (Step 4.6: Verify Angular Positioning)

Positions are evidence-based, consistent, distinct, and reference-anchored. **Diversity test:** mean pairwise Euclidean distance in (α , β , γ); if all derivatives sit within a sphere of radius <0.2, diversity is low — reconsider the extraction or document the source as unlayered. Target mean pairwise distance >0.3.

Mapping 6.16 (Step 4.7: Verify Synergistic Problem-Solving Potential)

Documented synergies are: explicit (recorded in the related derivatives); actionable (sufficient interface information); non-trivial (value beyond concatenation); traceable to source (not invented); **conditioned on reference (cite the registered RC-n / KC-n)**; and angularly grounded (mechanism explained by complementarity). Partners cited by `{uid:T8}`.

Mapping 6.17 (Step 4.8: Verify Completeness of Relevance to Reference Layer) — guiding and central

All derivatives must be built with respect to the reference layer. Content irrelevant to the reference layer disqualifies it from presence in a derivative. Derivatives must be traceable to the source, but this cannot waive relevance to the reference. **V3 strengthening:** every Relevance section must cite named register entries; a derivative whose relevance cites no RC-n / RD-n / KC-n / C-n fails this step regardless of traceability.

Mapping 6.18 (Step 4.9: Constraint Compliance Verification) — NEW IN V3

The Constraint Compliance Table (§8.9) is complete: every derivative × every Constraint Ledger entry carries a status — **Honors** (consistent, with evidence), **Tightens** (strengthens

the constraint), **Violates** (conflicts — must be flagged as a finding or resolved by boundary adjustment), or **Silent** (no contact — must be justified or feeds a blind spot / void). No unexplained Silent cells. An extraction that covers all challenges while leaving the ledger untested repeats the v0.2 defect this step exists to prevent.

Mapping 6.19 (Step 4.10: Blind Spot Audit) — NEW IN V3

Every derivative carries at least one grounded blind spot (a derivative with no identifiable blind spot signals ungrounded analysis, not perfection); every BS-n is grounded in Definition 6.1's admissible sources; every BS-n carries a coverage disposition per Definition 6.2; the set-level Blind Spot & Coverage Matrix (§8.12) is complete; uncovered blind spots are promoted to the Void Map.

Mapping 6.20 (Step 4.11: Void and Seed Verification) — NEW IN V3

Every void is grounded — traceable to an unengaged register entry (demand), an undeveloped source gesture (supply), or a DAG homology mismatch (structural). Voids that cannot be so traced are removed (§9.20). Every seed is actionable: it states what would have to be written or acquired, which register entries it serves, and a candidate angular position. Seeds are unminted unless deliberately chartered (UNID A5).

Mapping 6.21 (Step 4.12: Identity Verification) — NEW IN V3

- Every derivative carries its full anchor and tail-8 short anchor; kind: concept.
- All cross-links and functional dependencies use {uid} syntax and resolve within the report.
- All visas are complete (all nine minimal-visa fields) and well-formed (UUIDv7, variant bits).
- Tail-8 uniqueness holds within the report; collisions widened to tail-12 (UNID §4.3).
- Machine surfaces (YAML/JSON exports) carry full anchors only (UNID §5.3).
- The Registry Submission Block (§8.14) validates against the UNID minimal-visa schema; no record is represented as a registry resident before steward append.

Mapping 6.22 (Step 4.13: Reference-Side Verification) — NEW IN V3

- **Condensation fidelity:** sampled anchor-component phrases trace to the raw user-supplied reference (the reference-side mirror of Step 4.3).
- **CIM completeness:** every reference component has its operator bound and its output obligation discharged; no inert components.
- **Reference DAG:** acyclic, at component granularity, included in the output (§8.6), with homology notes against the derivative DAG.

- **Dominance:** ρ_{raw} and ρ_{op} reported; band classified; disposition applied (Definition 3.3).

7 Common Ground

Some content belongs to no specific concept — it applies to the entire text: framing (“This document proposes...”), meta-explanation (“The key point is...”), overarching claims affecting all concepts, universal dependencies, and reference-layer definitions when the source itself defines the problem context.

Management options: (1) extract separately as a Common Ground section preceding all derivatives — recommended for clarity; (2) repeat in each derivative (redundancy, but standalone completeness); (3) distribute to the most relevant concept with a cross-applicability note.

Angular position: usually at or near the origin ($\alpha, \beta, \gamma \approx 0,0,0$) — direct, concrete, universal framing adjacent to the reference layer. **V3 note:** material that is connective rather than common — joining specific concepts rather than applying to all — belongs in the Connective Tissue Ledger (Step 3.3), not in Common Ground.

8 Output Format

8.1 Output Format Purpose

The output format must simultaneously serve three needs: **(1)** traceability of each component to the source text; **(2)** readiness for evaluation and combination of derivatives relative to a reference layer; **(3)** — new in V3 — identity-anchored addressability, so every output element can enter the knowledge graph and survive recursive extraction passes without losing track.

8.2 Output Header

Required fields:

- **Extraction Title** and **protocol version** (this document's `<uid>` and version 0.3).
- **Source:** title/identifier + **UNID** `<uid:T8>` — minted at Step 0.1 if unregistered.
- **Reference Layer:** format used + **UNID** of the raw reference (if a registered or minted entity).
- **Word counts:** source; raw reference; operative reference.
- **Mixing ratio:** `p_raw` and `p_op` with dominance band and disposition (§3.7).
- **Entanglement density:** Low / Medium / High.
- **Identified conceptual units:** N, with the concept list — each name followed by its `<uid:T8>`.

8.3 Reference Layer

Insert the reference-layer specification per §3.2, followed by: the condensation provenance note (which raw passages constitute the anchor component); the CIM instantiation table (component → operator → output obligation, per §3.5); the RC / KC register; the Constraint Ledger; and the dimension register where applicable.

8.4 Entanglement Map (Source)

Minimum output: concept intersection points (sentence/paragraph/section); link type (causal, definitional, analogical, normative, methodological); entanglement intensity (low/medium/high).

8.5 Dependency Diagram (Source Derivatives)

Display functional dependencies of derivatives as a DAG or list, $A \rightarrow B$ (A depends on B for completeness), with nodes carrying `<uid:T8>` anchors. Report maximum dependency depth.

8.6 Reference-Side Entanglement Map and Component DAG — NEW IN V3

The same two artifacts for the reference layer at component granularity (§3.6): intersections among components and registered entries; the component dependency DAG; and **homology notes** — which reference-side edges are mirrored by derivative-side edges, and which are unmatched (candidate structural voids, cross-referenced to §8.13).

8.7 Angular Positioning Table

Derivative ($\langle \text{uid:T8} \rangle$)	Abstraction (α)	Analogical Distance (β)	Domain-Specificity (γ)
D1 ($\langle \text{uid:}\dots \rangle$)	0.3	0.1	0.7
D2 ($\langle \text{uid:}\dots \rangle$)	0.7	0.2	0.4
D3 ($\langle \text{uid:}\dots \rangle$)	0.5	0.8	0.3

Example structure. Each score carries a rationale in the derivative's own section.

8.8 Synergistic Problem-Solving Matrix

Pairwise matrix (upper triangle) of reference-conditional synergies. Every populated cell states the angular pattern, the mechanism, and **the RC-n / KC-n addressed**. The matrix is emitted both as a table and as structured data (machine surface with full anchors), in anticipation of the V4 synergy-combinatorics engine (§16.3).

8.9 Constraint Compliance Table — NEW IN V3

Derivative	C-1	C-2	C-3	...C-m
D1 ($\langle \text{uid:}\dots \rangle$)	Honors	Silent*	Honors	—
D2 ($\langle \text{uid:}\dots \rangle$)	Honors	Violates!	Tightens	—
D3 ($\langle \text{uid:}\dots \rangle$)	Silent*	Honors	Honors	—

Example structure. Statuses: Honors / Tightens / Violates / Silent. Every Violates cell is a flagged finding; every Silent cell (*) is justified or feeds a blind spot / void. Evidence pointers accompany each non-Silent cell.

8.10 Common Ground

Content applying to all derivatives; angular position $\approx (0,0,0)$, adjacent to the reference layer.

8.11 Writing Template for Each Derivative

Derivative: [Concept Name] — ($\langle \text{uid:T8} \rangle$)
 Anchor: [full 36-character UUIDv7] Kind: concept

Angular Position: (α, β, γ) = ([0,1], [0,1], [0,1])
 Rationale: [justification for the three axes]

Claims: [list of claims]
 Mechanism: [description of mechanism]
 Instances: [examples, each: " $\leftarrow$ Illustrates: claim/mechanism"]
 Boundaries: [limitations / caveats]

Functional Dependencies: [$\langle \text{uid:T8} \rangle \rightarrow \langle \text{uid:T8} \rangle$ list]
 Relevance to Reference Layer: [cites RC-n / RD-n / KC-n by name]

Constraint Compliance: [C-1: status+evidence; C-2: ...]

Blind Spots:

BS-1: [limitation] (grounds: Def. 6.1 class)

BS-2: ...

Coverage Means:

BS-1 → [sibling {uid:T8} | synergy | void V-n | external | irreducible]

Synergistic Problem-Solving Potentials:

- [Angular pattern] with {uid:T8};

Mechanism: [how they combine]

Addresses: [RC-n]

Alone, neither addresses: [what each misses alone]

Cross-links: [{uid:T8} references]

8.12 Blind Spot & Coverage Matrix — NEW IN V3

Set-level assembly of all BS-n entries: rows = blind spots (with owning derivative {uid:T8} and grounds class); columns = coverage disposition (sibling / synergy / void / external / irreducible) with the covering instrument named. Uncovered rows cite the Void Map entry they generate. The matrix answers the set-level question Step 4.10 verifies: **does the derivative set cover its own blind spots, and where it cannot, is the gap converted into a seed?**

8.13 Void Map & Generative Seeds — NEW IN V3

Void	Class	Description (grounding)	Generative seed
V-1	Demand	RC-n / C-n / RD-n no derivative or synergy engages (cite register entry)	What must be written/acquired; register entries served; candidate (α , β , γ); candidate handle (unminted)
V-2	Supply	Source gesture never developed (cite source location)	—
V-3	Structural	Unmatched reference-DAG edge / broken chain / uncovered blind spot (cite §8.6 / §8.12)	—

The seed catalogue is the project-level discovery frontier: input to subsequent extraction passes, recursive reference layers (V4), and the Missing Link Predictor (§16.3).

8.14 Registry Submission Block — NEW IN V3

A single fenced YAML fragment collecting every visa minted during the run: source and reference artifacts (Step 0.1) and all derivative concepts (Step 1.2), each with full anchors and complete minimal-visa fields. The block is formatted for direct append to registry.yaml **by the steward** after validation (unid_validate.py, checks V1–V10). Until appended, records are

minted-but-pending; the report **MUST** say so. Direct agent appends arrive with V5 governance (§14.3).

8.15 Verification Summary

Per-derivative check table — columns: Connection, Grounding, Binding, Composition, Independence, Angular Evidence, Synergies, **Constraints (4.9)**, **Blind Spots (4.10)**, **Identity (4.12)** — followed by the final checks:

- **Traceability sensitivity:** [X/Y] sampled elements traced (Step 4.3) — and condensation fidelity on the reference side (Step 4.13).
- **Angular diversity:** mean pairwise distance = [value] (target >0.3).
- **Reference-layer coverage:** challenge register coverage **and** Constraint Ledger coverage (no unexplained Silent columns).
- **Blind-spot coverage ratio:** covered BS / total BS, with voids generated.
- **Void census:** count by class (demand / supply / structural) and seeds issued.
- **Mixing ratio:** ρ_{raw} , ρ_{op} , band, disposition applied.
- **Identity validation:** visas complete; {uid} links resolve; tail-8 unique; submission block schema-valid.

PART III

Quality Assurance

Compiled on June 11, 2026

9 Failure Modes and Diagnosis

Failure modes 9.1–9.13 are carried from v0.2; 9.14–9.21 are new in V3 and correspond one-for-one to the new machinery (constraints, dominance, anchor invariance, identity, blind spots, negative space).

Definition 9.1 (Type-Based Segmentation)

Indication: derivatives named “Claims”, “Examples”, “Mechanisms”, “Caveats”. **Cause:** cuts internal connections; instances lose their principle, claims their grounding. **Remedy:** restart from Phase 1; identify concepts, not types.

Definition 9.2 (Re-Synthesis)

Diagnosis: output reads like new prose about the source; traceability audit fails. **Remedy:** use source language verbatim; only compress or cross-reference, never invent.

Definition 9.3 (Isolated Components)

Diagnosis: instances without principles, claims without mechanisms, boundaries without claims. **Remedy:** return to Phase 2; co-locate all four components.

Definition 9.4 (Over-Segmentation)

Indication: too many derivatives (>10 for a typical document); short, highly overlapping units; distinctness check fails (>30% shared content). **Remedy:** merge related concepts — a conceptual unit should be significant enough to stand alone. Record merges as visa events; anchors are never deleted.

Definition 9.5 (Under-Segmentation)

Indication: very long, internally heterogeneous derivatives; identifiable sub-concepts within one unit. **Remedy:** return to Phase 1; identify concepts more carefully.

Definition 9.6 (Over-Connection)

Indication: extensive interdependencies; functional independence <60%. **Remedy:** include more prerequisite content within derivatives, or revise unit boundaries.

Definition 9.7 (Undocumented Dependencies)

Indication: derivatives refer to other derivatives' concepts without explicit documentation. **Remedy:** return to Phase 3; document all functional dependencies in {uid} syntax.

Definition 9.8 (False Synergies)

Indication: documented synergies not grounded in the source. **Remedy:** remove invented synergies; document only those explicit or strongly implicit in the source.

Definition 9.9 (Missed Synergies)

Indication: obvious combinatorial opportunities undocumented. **Remedy:** systematically review all derivative pairs, especially angular complements.

Definition 9.10 (Angular Clustering)

Indication: mean pairwise distance < 0.2 . **Remedy:** either the source lacks layering (document it) or extraction missed layers — return to Phase 1.

Definition 9.11 (Ungrounded Angular Scores)

Indication: (α , β , γ) assigned with vague rationale. **Remedy:** return to Phase 2; assign positioning evidence, or note ambiguity instead of guessing.

Definition 9.12 (Disconnection from Reference Layer)

Indication: generic or empty “Relevance to Reference Layer” sections. **Remedy:** return to Phase 2 and clarify the reference specification; if the reference is too distant from the source, revise or re-select it.

Definition 9.13 (Unconditioned Synergies)

Indication: synergies documented without a tie to specific reference challenges. **Remedy:** state for each synergy the registered RC-n / KC-n it addresses that no derivative addresses alone; remove synergies that cannot be conditioned.

New failure modes in V3:**Definition 9.14 (Constraint Blindness — NEW)**

Indication: challenges covered (RC register fully addressed) while the Constraint Ledger is untested; Constraint Compliance Table absent or dominated by unexplained Silent cells. This is the defect observed in the v0.2-era SSB $\times$ ASM/CQR run: six RCs covered, zero constraints addressed. **Remedy:** apply Constraint Synthesis (Definition 3.2) and Step 4.9; re-test every derivative against every C-n.

Definition 9.15 (Reference Dominance (Mixing Failure))

Indication: $\rho_{op} > 0.5$; the derivative set mirrors the reference's taxonomy rather than the source's conceptual structure; extraction has become reference-driven cherry-picking (counter-pattern of §3.4). **Remedy:** apply the Dominance Guard — re-condense the reference, quote sparingly outside the anchor component, link the raw reference by `<uid>`, and re-run identification.

Definition 9.16 (Source Dominance (Starved Reference))

Indication: $\rho_{op} < 0.02$; relevance sections generic; the RC register nearly empty; derivatives unconditioned. **Remedy:** enrich the reference specification through the CIM (mint more challenges, constraints, dimensions from the raw reference), or reconsider whether the protocol applies.

Definition 9.17 (Anchor Drift)

Indication: the anchor component (Problem Domain / Context / Core Concept) is an agent paraphrase rather than a condensation; condensation-fidelity sampling fails; the user's intended reference content is no longer literally present. **Remedy:** re-condense by deletion/compression only (§3.3); re-anchor the angular origin; re-check β scores, which were measured against the drifted origin.

Definition 9.18 (Identity Defects)

Indication: unminted derivatives; cross-links by bare concept name instead of `<uid>`; incomplete visas; tail-8 collisions left unwidened; minted-but-pending records represented as registry residents. **Remedy:** Step 4.12; mint at extraction (Step 1.2), not retroactively; widen collisions per UNID §4.3; correct residency claims.

Definition 9.19 (Ungrounded Blind Spots)

Indication: blind spots written as free-floating criticism, traceable to none of Definition 6.1's admissible grounds (source boundaries, unengaged register entries, angular implications, dependency truncation). **Remedy:** Step 4.10 grounds test; delete what cannot be grounded — invented weakness is as much a fabrication as invented content.

Definition 9.20 (False Voids)

Indication: Void Map entries traceable to no unengaged register entry, no undeveloped source gesture, and no DAG mismatch. **Remedy:** Step 4.11; a void is a measured absence, not a wish list.

Definition 9.21 (Seed Anchor Fabrication)

Indication: UNID anchors minted for seeded entities that do not yet exist — a violation of UNID A5 (no fabricated identity; anchors are never positional). **Remedy:** seeds remain charters with candidate handles; mint on arrival, as immigrants are admitted.

10 Assessment Report Production Standards

10.1 Document Styling

10.1.1 Color Definitions

Name	RGB	Hex	Purpose
Dark Blue	(48,62,85)	#303E55	Primary (headers, lines)
Medium Blue	(79,140,183)	#4F8CB7	Secondary (sub-headers)
Light Blue	(227,242,253)	#E3F2FD	Background (definitions)
Blue-Gray	(241,245,249)	#F1F5F9	Background (patterns)

10.1.2 Status Color Coding

Status	RGB	Hex	Name
CLEAR / PASS / ADMITTED / HONORS	(34,139,34)	#228B22	Forest Green
BORDERLINE / UNCLEAR / TIGHTENS	(218,165,32)	#DAA520	Goldenrod
FAIL / REJECTED / VIOLATES	(220,20,60)	#DC143C	Crimson
PENDING / NOT EVALUATED / SILENT	(128,128,128)	#808080	Gray

V3 extends the status palette to the Constraint Compliance Table: Honors → Forest Green; Tightens → Goldenrod; Violates → Crimson; Silent → Gray.

10.1.3 Typography

Element	Font	Size
Body text	Arial	11 pt
H1 headers	Arial Bold	16 pt
H2 headers	Arial Bold	13 pt
H3 headers	Arial Bold	12 pt
Tables	Arial	11 pt (header row colored)
Identity strings (uid)	Monospace	9–10 pt

10.2 Table Formatting Requirements

Mandatory header row: every table includes a header row with column titles; no table is permitted without one. **Column widths:** text columns proportional to content (minimum 800 DXA); short label columns flexible; narrative columns 2000–3000 DXA; field/value tables [2500, 6860] DXA.

V3 table formats: the Constraint Compliance Table (§8.9) uses one status column per constraint with the §10.1.2 palette; the Blind Spot & Coverage Matrix (§8.12) carries the grounds class and the coverage instrument as named columns; the Void Map (§8.13) carries class and grounding columns. Identity columns (`{uid:T8}`) are typeset in monospace.

11 Validation Checklist

Before delivering output, verify:

- **Reference layer specified:** well-formed per §3.2 (editorial repair: v0.2 cited §2.2).
- **Anchor component invariant:** first component is a condensation of raw user reference content; fidelity sampled (§3.3, Step 4.13).
- **CIM instantiated:** every component bound to operators; no inert components; registers minted (RC / C / RD as applicable).
- **Constraint Ledger present:** explicit or synthesized (Definition 3.2) — or “none derivable” justified.
- **Mixing ratio reported:** ρ_{raw} , ρ_{op} , band, disposition (§3.7).
- **No type-based derivatives:** all units conceptual, not rhetorical.
- **Uniform composition:** claims + mechanisms + instances + boundaries (where source provides).
- **Connections intact:** instances with principles; claims with mechanisms; boundaries with claims.
- **Traceable:** every output element maps to a source location.
- **Distinct:** derivative overlap <20%, at documented entanglement points only.
- **Entanglement managed:** cross-links noted at separation points, in $\langle \text{uid} \rangle$ syntax.
- **Complete:** all source content assigned — derivative, common ground, or connective-tissue ledger.
- **Dependencies documented:** all functional dependencies explicit, $\langle \text{uid} \rangle$ syntax, DAG acyclic.
- **Reference-side maps present:** reference entanglement map + component DAG with homology notes (§8.6).
- **Independence sufficient:** all derivatives $\geq 60\%$.
- **Angular positions evidence-based:** every (α, β, γ) justified by content.
- **Angular diversity sufficient:** mean pairwise distance > 0.3 , or clustering documented as a source feature.
- **Synergies explicit, grounded, conditioned, angularly explained:** each cites its RC-n / KC-n and partners by $\langle \text{uid}:\text{T8} \rangle$.
- **Constraint compliance verified:** full table; every Violates flagged; every Silent justified (Step 4.9).
- **Blind spots grounded and covered:** every derivative ≥ 1 grounded BS-n; coverage matrix complete; uncovered spots promoted to voids (Step 4.10).
- **Voids grounded; seeds actionable:** Step 4.11; seeds unminted unless chartered.
- **Identity verified:** visas complete; anchors well-formed; tail-8 unique; machine surfaces full-anchor; submission block schema-valid (Step 4.12).
- **Relevance primacy holds:** no derivative whose relevance section fails to cite a named register entry (Step 4.8).

PART IV

Advanced Topics

Compiled on June 11, 2026

12 Invocation Syntax

INVOKE PROTOCOL "Extraction of Structured Derivatives from Intertwined Texts" v0.3
WITH REFERENCE LAYER specified according to [§3.2]

MODIFIERS (carried from v0.2):

- [] Pre-specify conceptual units: concepts
- [] Visual/tabular map of concept intersections: map entanglement
- [] Functional dependency graph: dependency graph
- [] 3D angular coordinate table: angular positioning chart
- [] Reference-conditional pairwise synergy analysis: matrix
- [] Extract common content separately: common ground
- [] Connection verification table: verify bindings
- [] Functional independence scores: assess independence
- [] Mean pairwise distance / diversity metrics: angular diversity assess
- [] Limit number of derivatives: N max derivatives
- [] Target output entanglement density: target density [Low/Med/High]
- [] Minimum functional independence: independence min X%
- [] Minimum mean pairwise angular distance: angular min distance X

MODIFIERS (new in V3; defaults marked):

- [x] Mint UNID anchors at extraction (Step 1.2): mint anchors (MUST; always on)
- [x] Component Interaction report (§3.5): cim report (default on)
- [x] Constraint Ledger + Compliance Table (§8.9): constraint ledger (default on)
- [x] Reference-side maps + homology (§8.6): reference dag (default on)
- [x] Blind spots + coverage matrix (§8.12): blind spots (default on)
- [x] Void map + generative seeds (§8.13): void map (default on)
- [x] Mixing ratio report (§3.7): mixing report (default on)
- [x] Registry Submission Block (§8.14): registry block (default on)
- [] Dry run (suppress registry block emission only): dry run

Note: identity-native invocation — passing {uid:sourceT8} and {uid:refT8} instead of file paths — arrives in V5 (§14.3). In V3 the inputs remain documents; their identities are resolved or minted at Step 0.1.

13 Advanced: Nested Extraction and Local References

For very long or complex texts, apply this protocol hierarchically: **first pass** extracts main conceptual clusters (3–5 high-level concepts); **second pass** extracts sub-concepts within each cluster using the same protocol; output is a tree of derivatives. Use for documents exceeding 5,000 words with 10+ distinct concepts.

Dependency propagation: dependencies between sub-derivatives within a cluster remain internal; cross-cluster dependencies propagate to the higher level.

Local references (formalized in V3, completed in V4): in nested extraction, each parent cluster serves as the **local reference layer** for its sub-derivatives — its condensed gloss is the local ORIGIN, and sub-derivative coordinates are measured against it. Absolute coordinates are recovered by composing local coordinates with the parent cluster's coordinates. V3 requires local references to be annotated as such; V4 makes them UNID-resolvable entities so that a sub-derivative's lineage (“positioned relative to {uid:cluster}, which is positioned relative to {uid:root-reference}”) is machine-traceable (§16.3, Nested & Local References).

Synergy at multiple levels: within clusters (between sub-derivatives) and between clusters (at the high level). Document both.

14 Identity Integration Roadmap (V3 → V5)

UNID — Unified Notation & Identity Discipline {uid:be1404ae} — guarantees known identity for every entity: identity precedes naming; the registry is a membrane; memory is append-only; names are credentials; no fabricated identity. This protocol adopts UNID in **three phases**, transitioning from isolated flat files to identity-anchored, addressable nodes within a continuous knowledge graph. V3 initiates the adoption; full adoption is reached at V5.

Phase	Name	Objective	Standing
V3	Foundational Identity Layer (Seeding)	Every derivative born with a permanent identity; relationships written in {uid} syntax.	Normative — in force in this document
V4	Graph-Wide Identity Propagation (Wiring)	Relationships, synergies, and artifacts become entities; recursive reference layers.	Planned — requires UNID minor-version amendment
V5	Identity-Native Operation (Autonomous Graph)	The protocol operates on the registry and graph directly; full adoption.	Planned — requires governance gates

14.1 V3 — Foundational Identity Layer (The Seeding Phase) — normative

Objective: equip the protocol to generate UNID-compliant anchors at the point of derivative creation, ensuring every extracted concept is born with a permanent identity.

1. **Minting at extraction.** The moment the agent identifies a valid conceptual unit (Step 1.1) and it passes the structural filters, the agent **MUST** instantly mint a UNID anchor for it, auto-generating the minimal visa alongside the derivative's structural components. The agent **MUST** assign **kind: concept** to every extracted derivative (Step 1.2).
2. **Output header modification.** The Output Header **MUST** include the UNID of the source document and the UNID of the reference layer (minted at Step 0.1 if unregistered). Each derivative's writing template **MUST** include its full anchor and tail-8 short anchor (§8.2, §8.11).
3. **Cross-linking via identity.** Cross-links and functional dependencies within derivatives **MUST** use UNID syntax — {uid:T8} — instead of bare concept names, establishing the first level of machine-readable relationships (§8.5, §8.11).

V3 discipline (what V3 deliberately does not do): no kinds outside the closed UNID taxonomy are minted (synergies remain text citing participant {uid}s); no direct registry appends (the Registry Submission Block is steward-validated); seeds remain unminted charters (UNID A5).

14.2 V4 — Graph-Wide Identity Propagation (The Wiring Phase) — planned

Objective: move beyond assigning IDs to individual derivatives; track relationships, synergies, and the evolution of concepts in the UNID structure.

1. **Identity for the interstitial (synergies as entities).** A documented synergistic problem-solving potential between two or more derivatives ceases to be a paragraph of text: the agent **MUST** mint a UNID for each verified synergy, assigning a new kind (kind: synergy or kind: relationship) — requiring a **minor-version amendment to the UNID kind taxonomy** per UNID §13. Benefit: future analyses can cite the combination of ideas directly rather than re-deriving the connection.
2. **Identity for the artifacts.** The entanglement maps, dependency DAGs, and the synergy matrix generated by an extraction **MUST** be minted as separate entities (kind: artifact), bound to the parent extraction report via their metadata.
3. **Recursive reference layers.** A previously extracted derivative (kind: concept) or documented synergy (kind: synergy) can be explicitly fed back as a Conceptual Anchor reference layer for a new extraction run. Because the reference layer now has a UNID, the lineage of the new derivatives is perfectly traceable back to the original source through the reference node.

14.3 V5 — Full Identity-Native Protocol Operation (The Autonomous Graph) — planned

1. **Identity-native input/output.** INVOKE PROTOCOL is updated: instead of file paths, the user passes the UNID of the source (`{uid:sourceT8}`) and of the reference layer (`{uid:refT8}`). The protocol resolves both against the registry. The agent is authorized to perform direct, validated appends to `registry.yaml` — inserting newly minted derivatives and their relationship edges automatically, subject to the validator (V1–V10) and to human veto / adversarial review where required by UNID §6.3 governance.
2. **Continuous sub-system updating (homeostasis).** On each new extraction, the agent searches the UNID network for existing concepts overlapping the newly discovered ones. If a new derivative is functionally identical to an existing `{uid:concept}`, the agent **MUST** propose an **eviction by merge** (`superseded_by`) under UNID §6.3 — preserving the network's cleanliness without deleting history (anchors resolve forever; merge changes standing, never existence).
3. **Chain analysis realization.** Because every claim, concept, source, and synergy has a permanent UNID, an analyst — or an AI Missing Link Predictor — can write queries that traverse the graph: `*find all derivatives that depend on {uid:X} and were generated using reference layer {uid:Y}*`. The system traces the exact intellectual lineage of an idea across multiple recursive extraction passes without ever losing track.

14.4 Adoption Discipline

Phase advancement is gated, not assumed: V4 requires the UNID taxonomy amendment (proposal, adversarial review, objection window, human veto absolute — UNID §13) and

validator support for the new kinds; V5 requires the governance gates of UNID §6.3 wired into the append path. Each advancement is recorded as a version event on **this protocol's own anchor** (uid:e9fc1e1f) — per UNID §8, versions are events on one anchor, not new anchors. Identity standing, content admissibility (Ghost File Admissibility Protocol (uid:eee89af7)), and artifact versioning remain three independent axes throughout.

15 Relationship to Source Instructions

This protocol combines the **Decomposition Instruction** (separating based on conceptual core) and the **Entanglement Unraveling Instruction** (separating interwoven threads). Both are special cases: terminology differs (“fractions” / “threads” / “derivatives”) but the logic is identical — identify conceptual units; assess entanglement; assign content; assemble preserving connections; verify integrity.

The general protocol extends the source instructions with: (1) explicit functional dependencies; (2) reference-layer conditioning for synergy evaluation; (3) angular positioning in a 3D divergence space; (4) synergistic problem-solving potentials from angular complementarity; (5) standing review and purging for a self-healing knowledge base (0.2); (6) evidence-processing standards (0.2); (7) bias controls (0.2); (8) cross-assessor calibration (0.2); (9) fitness guidance for operational scalability (0.2); and — new in 0.3 — (10) component-resolved reference interaction (CIM); (11) identity seeding under UNID; (12) negative-space outputs (blind spots, voids, seeds, connective tissue); (13) dominance governance of the source–reference balance.

16 Extensions and Future Work

16.1 Quantitative Metrics

- **Entanglement coefficient:** fraction of sentences serving multiple concepts.
- **Connection completeness:** fraction of instances having their principle in the same derivative.
- **Compositional similarity:** KL divergence between component-type distributions across derivatives.
- **Independence score:** fraction of a derivative understandable without external reference.
- **Dependency depth:** maximum path length in the dependency DAG (lower is better).
- **Angular diversity:** mean pairwise Euclidean distance in (α, β, γ) .
- **Synergy coverage / density:** fraction of derivative pairs with documented synergy; average synergies per derivative.
- **Reference-layer alignment:** correlation between relevance scores and angular distance from origin.
- **Calibration coefficient / standing stability:** inter-assessor agreement; retained-to-challenged ratio (0.2).
- **Constraint coverage (NEW):** fraction of Constraint Ledger entries with at least one non-Silent derivative row.
- **Blind-spot coverage ratio (NEW):** covered BS-n / total BS-n.
- **Void density (NEW):** voids per derivative, by class; **seed yield:** seeds issued per extraction.
- **Mixing ratio (NEW):** ρ_{raw} , ρ_{op} and band history across runs.
- **Condensation fidelity (NEW):** fraction of anchor-component phrases tracing to raw reference content.
- **Homology index (NEW):** fraction of reference-side DAG edges mirrored by derivative-side edges.

16.2 Automated Tooling

The protocol is algorithmic and can be implemented as: an LLM-based extraction pipeline with dependency analyzer and angular positioning engine; a human-in-the-loop annotation tool with independence scoring and 3D visualization; a hybrid (LLM proposes, human verifies); graph-based dependency visualizers; a reference-conditional synergy matrix generator with composition interface; calibration tracking and standing-review workflow automation; bias diagnostics (all carried from 0.2). V3 adds: a UNID visa generator with tail-collision detection; a registry-submission validator (wrapping `unid_validate.py`); a CIM instantiation assistant; and mixing-ratio instrumentation.

16.3 Candidate Engines — dispositioned against the roadmap

Engine	Phase	Disposition and V3 hook
Auto-pilot Discovery Engine	V4–V5	Autonomous selection of next sources/references to extract, driven by the seed catalogue and registry graph. V3 hook: machine-readable Generative Seeds (§8.13).
Automated Synergy Combinatorics Engine	V4	Systematic enumeration and testing of pair/triplet combinations against the RC register; requires synergies as entities (kind: synergy). V3 hook: synergy matrix emitted as structured data with full anchors (§8.8).
Missing Link Prediction	V4–V5	Predicts absent nodes/edges from graph structure; the designated consumer of the Void Map and homology index. V3 hook: voids and unmatched DAG edges as labeled training targets (§8.6, §8.13).
Nested & Local References	V3 (partial) → V4	Parent clusters as local reference layers — formalized in §13; V4 makes local references UNID-resolvable for machine-traceable nested lineage.
Dynamic Boundary Constraint Tuning	V5	Runtime adjustment of extraction thresholds (independence minima, angular targets, density bands) from observed yield. V3 hook: the Constraint Ledger records which constraints bind (active) versus stay slack — the observability tuning requires.

16.4 Domain Adaptations

Domain	Concepts	Dependencies / synergies	Angular axes (α / β / γ)
Legal documents	Legal principles, doctrines	Precedent relationships; combined defenses, multi-jurisdictional arguments	Doctrinal–abstract / direct–precedent–analogical / statutory–customary
Scientific articles	Hypotheses, methods, results	Theoretical foundations; cross-experimental validations	Empirical–theoretical / observation–model inference / sub- vs cross-disciplinary
Code documentation	Modules, APIs, algorithms	Input/output diagrams; compositional patterns	Implementation–specification / direct use–design pattern / language-specific–portable

Domain	Concepts	Dependencies / synergies	Angular axes (α / β / γ)
Process engineering	Unit operations, control strategies	Material/energy flows; process integration	Equipment—principles / direct—analogue process / process-specific—general

16.5 Optimization Frameworks

- Reformulate extraction as constrained optimization: minimize total dependency weight while maintaining connection integrity.
- Bi-objective: maximize functional independence and reference-conditional synergistic potential; explore the Pareto frontier across extraction configurations.
- Maximize angular diversity subject to traceability and connection constraints.
- Maximize reference-layer coverage — in V3 this objective explicitly includes Constraint Ledger coverage and blind-spot self-coverage, not challenge coverage alone.
- Minimize uncovered blind spots subject to the no-fabrication constraints (Definitions 6.1, 9.19).

16.6 Toward Rigorous Topology Construction — initiated in V3

V3 initiates a gradual movement toward equipping the framework with a rigorously defined topology, against the following definition, which is adopted as the target standard:

Target Definition (Topology)

A topology on a set X is a distinguished collection of subsets of X that contains the extremes, is closed under arbitrary joins (unions), and finite meets (intersections), thereby specifying which regions are locally observable and enabling the notions of continuity, convergence, connectedness, compactness, and neighborhood without requiring any metric, geometric, differentiable, or algebraic structure.

16.6.1 Where V3 Diverges from the Definition

1. **Metric dependence.** V3's angular machinery is a pointed metric space: positions are points in $[0,1]^3$, the reference at the origin, and the working analytics (diversity, adjacency, opposition) are Euclidean-distance statements. The target definition requires none of that structure — and pointedly does without it. V3's analytics are therefore finer-grained but more arbitrary: a score of $\alpha = 0.55$ versus 0.60 carries pseudo-precision that no verification step can fully ground.
2. **Points, not regions.** The carrier set X is the derivative set (extended in V4 by reference components and synergies). V3 observes X through individual points — single derivatives — whereas a topology observes through **regions** (open sets, “which regions are locally

observable”). The closest V3 comes to region-thinking is the functional-independence score: a derivative readable standalone is, implicitly, a observation-closed singleton.

3. **No closure discipline.** V3's distinguished subsets — synergy pairs and triplets, DAG ancestor closures, entanglement clusters, RC-coverage sets, common ground — form at best a **subbasis**: the collection is not closed under arbitrary unions (the union of two documented synergies is not itself a documented structure) nor under finite intersections (the overlap of two clusters is not a recognized unit), and the extremes are present only incidentally (X as “everything”, the empty set trivially).
4. **Finite carrier.** X is finite. On a finite set, topologies correspond exactly to preorders (the Alexandrov correspondence): open sets are the down-sets of a specialization preorder, and the discrete topology — every subset open — trivializes all the target notions. The construction therefore has content only if the preorder is chosen meaningfully.

16.6.2 The Natural Construction V3 Already Carries

The meaningful preorder is already in the output: the **functional-dependency DAG**. Take its reflexive–transitive closure as the specialization preorder and declare a subset of X **open iff it is dependency-closed** — with every derivative it contains, it contains everything that derivative depends on. Then:

- **Open set = self-contained bundle.** A dependency-closed set is exactly a set of derivatives readable with no external reference — the topological formalization of Definition 2.5's functional independence. “Locally observable region” acquires the precise meaning: a bundle you can hand to a reader whole.
- **Extremes and closure hold by construction:** $\emptyset$ and X are dependency-closed; arbitrary unions and arbitrary (hence finite) intersections of dependency-closed sets are dependency-closed. The collection is a genuine (Alexandrov) topology, not a subbasis.
- **Continuity.** A map between extractions — a recursive pass, a re-anchoring to a new reference layer, a merge — is continuous iff preimages of self-contained bundles are self-contained: it breaks no prerequisite chains. “Safe transformation of a knowledge base” becomes a checkable property.
- **Convergence.** Recursive extraction passes converge when the lattice of open sets stabilizes — successive passes stop discovering new dependency structure.
- **Connectedness.** X is connected iff it does not split into two nonempty dependency-closed parts with no cross-links: no isolated knowledge islands. This is the precise form of the intuition that **continuity (connectedness) of derivatives is the most value-producing outlook** — connected derivative sets are the ones whose lineage compounds.
- **Compactness.** On finite X compactness is automatic and therefore uninformative; it acquires content only as X grows toward the registry-wide graph (V5) or toward infinite refinement limits. This is stated honestly rather than claimed as a benefit.

16.6.3 Ramifications — Benefits and Losses of Analytic Power

The move trades metric analytics for structural guarantees. The central tension: **continuity/connectedness of derivatives** as the most value-producing outlook, versus **scattering the units to cast a wider net** — increasing the discovery field radius.

Benefits (topological regime)	Losses of analytic power
Lineage integrity: continuous maps preserve dependency closure, so chain analysis (§14.3) survives every transformation.	The diversity discipline (mean pairwise distance >0.3) has no topological counterpart; discovery pressure must be re-expressed (e.g., open-cover cardinality, anti-chain width).
Stability statements: re-anchoring to a nearby reference induces a controlled deformation of the derivative set — perturbation tolerance becomes provable rather than observed.	Adjacent vs. opposing synergy patterns collapse: Alexandrov topology sees order, not distance; “maximal contrast” (the Opposing pattern) is invisible to it.
Compositional reasoning: unions of self-contained bundles are self-contained — reuse closes under joins by construction.	The three angular semantics (abstraction, analogy, domain) are not recoverable from open sets alone; the interpretive richness of (α, β, γ) is lost unless carried as auxiliary structure.
Homology with the reference-side DAG becomes functorial rather than ad hoc — matched structure is preserved by continuous maps.	Enforcing connectedness suppresses precisely the high-distance outliers that maximize the discovery field radius; a connected-by-construction set biases toward incremental adjacency and shrinks the explored design space.
Removes pseudo-precision: qualitative regions replace contestable point scores; verification burdens drop.	Quantitative gates (diversity thresholds, distance minima) disappear; QA must be rebuilt on structural criteria.

Resolution — the dual regime. The trade-off is not to be decided once for the whole framework. The recommended operating point is: maintain a **connected core** — dependency-closed, $\{\text{uid}\}$ -chained, topology-governed, where continuity of derivatives produces compounding value — and a **scattered frontier** — deliberately disconnected probes at large angular distance, casting the wide net. The Void Map and Generative Seeds (§8.13) are the designed conversion mechanism between the two: scattering is a discovery investment whose value is realized when a frontier unit acquires a dependency path into the core, and the topology makes that conversion **measurable** (a frontier unit joins the core exactly when its singleton ceases to be open in isolation and enters a larger self-contained bundle).

Phasing. V3 documents the divergences and the construction (this section) while keeping the metric machinery normative. V4 pilots dual bookkeeping — Alexandrov open-set reporting alongside the angular tables — on at least two corpora. V5 decides primacy on observed analytic yield, by the same governed amendment path as every other structural change (§14.4).

17 Version History

17.1 Current Version (0.3 — designated V3)

Execution Date	2026-06-11
Review Cycle	Quarterly
Identity	Version event on anchor {uid:e9fc1e1f} (UNID §8: versions are events, not new anchors)

Changes from Version 0.2:

Section	Type	Summary
§1, §2.6–2.9	New	Entropy-reduction framing; Relevance Primacy elevated to a universal principle; Identity at Birth; Conditioning-not-Mixing; Negative Space as output.
§3.3	New	Anchor Component Invariant: first component of every format (Problem Domain / Context / Core Concept) is an imposed condensation of raw user reference content; immune to extreme transformations; other components by agent judgment.
§3.5	New	Component Interaction Model: per-component operators (ORIGIN, SCOPE, TARGET, GATE, ORACLE, LENS, BASIS, FRAME, SPACE), phase bindings, and output obligations for all components of all three formats; Constraint Synthesis Rule (Definition 3.2) closing the constraints gap observed in the SSB × ASM/CQR run.
§3.6, §8.6	New	Reference-side entanglement map and component dependency DAG, with homology analysis against the derivative DAG; included in the final analysis.
§3.7	New	Mixing ratio (p_{raw} , p_{op}), dominance bands, and the Dominance Guard, replacing reliance on user discretion for source–reference size balance.
§6.1–6.5	Updated	Phase 0 restructured (CIM instantiation, condensation, dominance); Step 1.2 Mint at Extraction; assignment rules 6–7 (constraint and blind-spot evidence); Step 3.3 Negative Space Assembly; verification steps 4.9–4.13.
§8	Updated	Output header carries source/reference UNIDs and mixing ratios; new sections 8.6, 8.9, 8.12, 8.13, 8.14; derivative template extended with anchors, constraint compliance, blind spots, coverage means.
§9.14–9.21	New	Failure modes: constraint blindness, reference/source dominance, anchor drift, identity defects, ungrounded

Section	Type	Summary
		blind spots, false voids, seed-anchor fabrication.
§11	Updated	Validation checklist extended to the V3 machinery; cross-reference repairs (v0.2's §2.2/§3.5 orphans).
§12	Updated	V3 modifiers; dry-run flag; identity-native invocation deferred to V5.
§13	Updated	Local references formalized (parent cluster as local ORIGIN).
§14	New	Identity Integration Roadmap: V3 Seeding (normative) → V4 Wiring → V5 Autonomous Graph; adoption gates; eviction-by-merge homeostasis; chain analysis realization.
§16.3	New	Candidate engines dispositioned: Auto-pilot Discovery, Synergy Combinatorics, Missing Link Prediction, Nested & Local References, Dynamic Boundary Constraint Tuning.
§16.6	New	Topology program: target definition adopted; V3 divergences enumerated; Alexandrov/dependency-closure construction; benefits vs. losses (connectedness vs. discovery radius); dual-regime recommendation; V4/V5 phasing.

17.2 Version 0.2 (superseded)

Execution date 2026-02-01; quarterly review. Changes from 0.1: §3.4 new; §3.5 improved; §4.5 new; §9 updated; updates throughout.

— End of Protocol 0.3 —